INCH-POUND

MIL-S-52098C
8 September 1992
SUPERSEDING
MIL-S-52098B
2 August 1971

MILITARY SPECIFICATION

STIFFENER ASSEMBLY, BALK CONNECTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the stiffener assembly and its components for support of the deck balk superstructure.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-T-704	- Treatment and Painting of Material.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking for US Military Property.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DRAWINGS

ME

TA13208E9871	- Stiffener Assembly, Balk Connecting.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development and Engineering Center. ATTN: SATBE-JBS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues

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of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953	- Strapping, Flat Steel and Seals.
D 4675	- Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

A5.10	- Bare Aluminum and Aluminum-Alloy Welding Electrodes and Rods.
D1.2	- Structural Welding Code-Aluminum.

(Application for copies should be addressed to the American Welding Society, 555 NW 42nd Street, Miami, FL 33126.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The stiffener assembly and its components shall be as shown on Top Assembly TA13208E9871 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly and operation of the stiffener assembly. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2) a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The stiffener assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the stiffener assembly may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the stiffener assembly may be newly fabricated from recovered materials to the maximum

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extent practicable, provided the stiffener assembly produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the stiffener assembly.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A5.10, classification ER4043, size as required.

3.5 Identification marking. The stiffener assembly or components shall be identified in accordance with MIL-STD-130 and shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.6 Treatment and painting. The stiffener assembly or components, as applicable, except pins and screw threads shall be cleaned, treated, and painted in accordance with MIL-T-704, type B, color number 34088 in accordance with FED-STD-595. Contact surfaces, regardless of whether such contact is between similar or dissimilar metals, shall be cleaned, treated, and painted prior to assembly.

3.7 Stenciling. The nomenclature, Federal stock number, weight, and cube shall be stenciled on each stiffener assembly. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. The markings shall be Gothic capitals and Arabic numerals, .750-inch high. Stenciling shall conform to MIL-T-704.

3.8 Workmanship.

3.8.1 Aluminum fabrication. Before any work or layout is done, aluminum shall be straightened by methods that will not result in injury to the metal. Care shall be used in handling aluminum during fabrication to avoid excessive scratching or damage to edges. Finished members shall be free from laminations, kinks, bends, scratches, or winds. Corners shall be square and true. Aluminum .375-inch thick and over shall be cut by routing or sawing. Aluminum under .375-inch thick may be cut by shearing. Aluminum shall not be cut by methods using heat.

3.8.2 Riveting. Components shall be bolted in position before riveting. Jigs or frames shall be used to maintain alignment and tolerance. Holes for countersunk heads shall be of such depth and angle that the head shall fill the hole and be flush with the surrounding surface. Heads of driven rivets shall be uniform in shape and concentric with the rivet shank axis, and shall have full bearing on the surfaces of the joined members. Rivets shall fill the holes in the joined members. Caulking or redriving of rivet heads shall not be permitted. Rivet tension shall not be created by driving in incorrectly fitted members. A rivet with a head deficient in size, not uniform in shape, off-center, loose, or misshapen in any way shall be considered defective. Defective rivets shall be removed without injury to the joined metal. Rivets shall be driven cold.

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3.8.3 Welders and welding.

3.8.3.1 Welders or welding operator. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either AWS D 1.2 or the ASME code for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined. Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of poor welds, the Government reserves the right to require retesting of any welder or welding operator (see 6.5).

3.8.3.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welds shall be done by the shielded metal-arc, gas metal-arc (MIG), or gas tungsten-arc (TIG) process. Welding procedures shall be qualified in accordance with applicable ASME or AWS codes, and welding and welds shall be in accordance with ASME or AWS codes. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wirebrushed clean.

3.8.4 Interchangeability. Stiffener assemblies shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable.

3.9 Cadmium plated fasteners. No cadmium plated fasteners shall be used unless deemed absolutely necessary (see 6.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with

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all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.7.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article stiffener assembly or components(s) as applicable, shall be examined as specified in 4.4.2.1 and for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination.

4.4.2.1 Individual. Each stiffener assembly shall be inspected for conformance to the dimensions covered by the quality assurance provisions (QAP) shown on the drawings under "Major - 100 percent inspection". Nonconformance to any of these dimensions shall constitute failure of this examination and shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.5 Inspection procedure.

4.5.1 Examination. The stiffener assembly or component, as specified herein, shall be examined for the following defects:

Major

101. All dimensions designated major on the QAP other than those under 100 percent inspection not as specified (see 3.1).
102. Material not as specified (see 3.3).
103. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment as specified (see 3.3.1).
104. Dissimilar metals as specified in MIL-STD-889 are effectively insulated from each other as specified (see 3.3.1.1).
105. Contractor does not have documentation available for identification of material, material finishes, or treatments (see 3.3.1.2).
106. Components missing, not interchangeable or not as specified (see 3.1 and 3.8.4).
107. Workmanship not as specified (see 3.8).
108. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal (see 3.8.3.2).
109. Riveting not as specified (see 3.8.2).
110. Welders and welding operators not certified as specified (see 3.8.3.1).
111. Welding and welds not as specified (see 3.8.3.2).

Minor

201. All dimensions not designated major on the QAP not as specified (see 3.1).
202. Identification marking incorrect, missing or illegible (see 3.5).
203. Treatment and painting not as specified (see 3.6).
204. Stenciling not as specified (see 3.7).
205. Cadmium plating not as specified (see 3.9).

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105 by using table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

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4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

112.	Components and parts which are acquired separately from the stiffener not packed in a shipping container as specified for level A or B (see 5.1.1.1 and 5.1.2.1).
113.	Gross weight of wood boxes exceeds 200 pounds for level A or B (see 5.1.1.1 or 5.1.2.1).
114.	Strapping not as specified for level A (see 5.1.1.1).
115.	Gross weight or size of fiberboard boxes exceeds the limitations of the applicable box specification for level B (see 5.1.2.1).
116.	Marking missing, illegible, incorrect, or incomplete (see 5.2).

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C, as specified.

5.1.1 Level A. Each completely assembled (with pins in place) stiffener assembly shall be prepared for shipment loose.

5.1.1.1 Components and parts. When the pin, spring, and spring holder assembly are acquired separately from the stiffener assembly, the items shall be packed in close-fitting wood boxes conforming to PPP-B-601, overseas type, style optional. The gross weight of each box shall not exceed 200 pounds and the boxes shall be closed in accordance with the appendix to the box specification. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc coated, size as applicable and ASTM D 4675.

5.1.2 Level B. Each completely assembled (with pins in place) stiffener assembly shall be prepared for shipment loose.

5.1.2.1 Components and parts. When the pin, spring, and spring holder assembly are acquired separately from the stiffener assembly, the items shall be packed in close-fitting wood boxes conforming to PPP-B-601, domestic type, style optional; in close-fitting fiberboard boxes conforming to PPP-B-640, class 2, style optional; or PPP-B-636, V3c, V11c, V13c, or V15c, style optional. The gross weight of each wood box shall not exceed 200 pounds and the boxes shall be closed in accordance with the appendix to the box specification. Strapping shall be as specified in 5.1.1.1. The gross weight or size of fiberboard boxes shall not exceed the limitation of the applicable box specification. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall be as specified in 5.1.1.1.

5.1.3 Level C. Each completely assembled (with pins in place) stiffener assembly shall be prepared for shipment loose. Components and parts, when acquired separately from the stiffener assembly, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in fiberboard or wooden containers complying with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

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5.2 Marking. Shipping containers and loose assemblies shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The stiffener assembly is intended for use as a beam for support of the balk superstructure and as a means of connecting the balk superstructure to the supporting floats.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required for inspection and approval, and the number of units required (see 3.2 and 6.3).
c. Level of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item(s) should be a production model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Quality assurance provisions (QAP). The contracting officer should require the supplier to maintain records of all QAP inspections. A suggested paragraph is as follows:

> "The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the

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specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
3.8.3.1	DI-MISC-80876	Welding Procedure Qualification Test Report
3.8.3.1	DI-MISC-80875	Welding Procedures

6.6 Cadmium plated fasteners. The contracting officer should require the contractor to justify the use of cadmium plated fasteners in writing (see 3.9). Citing standard DoD or commercial practice will not be considered sufficient justification. A specific rationale is required for each specific/ different applicable of cadmium plated fasteners.

6.7 Definitions.

6.7.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.8 Subject term (key word) listing.

Balk connecting stiffener
Balk superstructure
Beam, support
Deck balk superstructure
Stiffener, balk connecting
Stiffener, balk superstructure
Stiffener, support

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 5420-0174

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
The submitter of this form must complete blocks 4, 5, 6, and 7.
The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-52098C	2. DOCUMENT DATE (YYMMDD) 920908

3. DOCUMENT TITLE
Stiffener Assembly, Balk Connecting

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Geraldine F. Harlow	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3469 (2) AUTOVON 654-3468
c. ADDRESS (Include Zip Code Army Belvoir RDE Center ATN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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